

Rossby Wave Ray-tracing: A Modern Approach

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2026403126

May 25, 2026

Abstract

1 Introduction

Rossby waves are large-scale waves that appear in stratified fluids on rotating bodies [2]. On earth, they appear both in the atmosphere and oceans and are almost perpendicular to the earth's axis of rotation. Atmospheric Rossby waves were first described in 1939 by Swedish-American meteorologist Carl-Gustaf Rossby.

This project will consider atmospheric Rossby waves and make use of modern numerical algorithms to analyze their behavior. Much of the work is based on the 1993 paper *Rossby Wave Propagation on a Realistic Longitudinally Varying Flow* by Hoskins and Ambrizzi [1]. This paper, however, differs in that different numerical methods for solving the problem will be compared.

2 Theory

This section serves as a brief introduction to the mathematical interpretations of Rossby waves, as well as some of the numerical methods used.

2.1 The Stationary Wavenumber K_s

Central to any theory dealing with waves is the concept of a wavenumber K which describes the spatial frequency of a wave. Its definition is simply the reciprocal of the wavelength λ , i.e. $K = \frac{1}{\lambda}$. Assuming that the wave is on the form $\exp[i(kx + ly)]$,

The central concept in the paper is the stationary wavenumber K_s . It tells us, at each location in the atmosphere, what wavelength a Rossby wave must have to remain stationary relative to the ground. The formula is:

$$K_s^2 = \frac{\beta_M - \partial^2 U / \partial y^2}{U} \quad (1)$$

Where U is the background zonal wind and β_M is the gradient of Earth's vorticity in Mercator coordinates.

When K_s is large and uniform over a region, that region acts as a waveguide: waves of the appropriate scale are trapped and can travel long distances. When K_s^2 becomes negative, waves cannot propagate at all (they are "evanescent").

2.2 Ray Tracing

Ray tracing treats a wave packet as a particle whose position (x, y) and wavenumber (k, l) evolve according to Hamilton's equations. The key idea is that waves bend toward regions of higher K_s , similar to how light bends in a medium with varying refractive index. By integrating these equations forward from a source location, we can predict where wave energy will go.

3 Data and Methods

3.1 Data

- Source data:
 - ERA5 reanalysis
 - Spatial variables: zonal (u) and meridional (v) winds at 300hPa
 - Time variable: Monthly from January 1940 to April 2026
 - Spatial resolution: $0.25^\circ \times 0.25^\circ$
- Used data:
 - Time-averaged over December-February (referred to as DJF)
 - Spatial resolution: $2.5^\circ \times 2.5^\circ$

3.2 Computing K_s

3.3 Ray Tracing Implementation

4 Results

4.1 Comparison with Hoskins & Ambrizzi (1993)

5 Discussion

References

- [1] Brian J. Hoskins and Tercio Ambrizzi. Rossby wave propagation on a realistic longitudinally varying flow. *Journal of Atmospheric Sciences*, 50(12):1661 – 1671, 1993.
- [2] National Oceanic and Atmospheric Administration. What is a rossby wave?, 2024.